

Enhancing Ethanol Coupling to Produce Higher Alcohols by Tuning H_2 Partial Pressure over a Copper-Hydroxyapatite Catalyst

Bai-Chuan Zhou, Wen-Cui Li, Wen-Lu Lv, Shi-Yu Xiang, Xin-Qian Gao, and An-Hui Lu*

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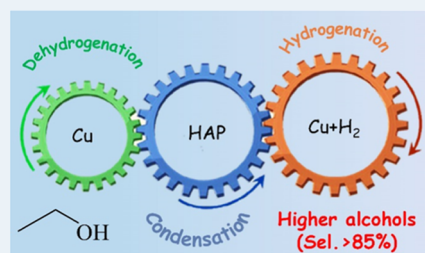
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ABSTRACT: Catalytic upgrading of ethanol, as a platform molecule from biomass to higher alcohols (C_{4-12}), is a low-carbon route for value-added chemical production. However, the products are generally obtained in low selectivity due to the uncontrollable reactivity of intermediates that cause a complex reaction network. In this study, we show that unsaturated intermediates of aldehydes can be rapidly hydrogenated by surface hydrogen species during the ethanol upgrading process, thereby greatly inhibiting the cyclization reaction of aldehydes. Specifically, the product distributions on the Cu-hydroxyapatite (Cu-HAP) catalyst shift stepwise to higher alcohols from aromatic oxygenates with the partial pressure of hydrogen increasing from 0 to 95 kPa. Kinetic measurements and in situ ethanol infrared results indicated that the intermediates during this process are acetaldehyde and 2-butenal. Combined with physical structure and chemical state analysis of the catalyst, we found that Cu sites catalyze the hydrogenation of the $C=C$ bond of 2-butenal under a hydrogen atmosphere. The $C-C$ coupling of ethanol to higher alcohols over Cu-HAP follows the Guerbet mechanism. In comparison, on bare HAP, *n*-butanol is formed as a primary product even though little amount of acetaldehyde was detected, indicating that ethanol proceeds mainly in a direct coupling process to yield higher alcohols. This study introduces an efficient ethanol valorization approach that is enabled by subtle control of the intermediate conversion over the Cu-HAP catalyst by the hydrogen partial pressure.

KEYWORDS: ethanol, higher alcohols, copper-hydroxyapatite, dehydrogenation, hydrogenation



1. INTRODUCTION

The catalytic conversion of renewable ethanol to desirable value-added products offers an alternative pathway toward reducing carbon emissions.^{1–3} Ethanol can be proceeded in $C-C$ coupling to synthesize valuable long-carbon-chain oxygenates, such as C_4-C_{12} aliphatic alcohols. Higher alcohols are valuable precursors for plasticizers and also serve as clean fuel additives because of high energy density and limited water miscibility.⁴ The above C_4-C_{12} higher alcohols are mainly manufactured industrially via the oxo process, which involves hydroformylation and subsequent hydrogenation of the resultant aldehydes.⁵ Therefore, the use of ethanol from biomass to produce C_4-C_{12} aliphatic alcohols meets the target of carbon neutrality and the green development policy.^{6–9}

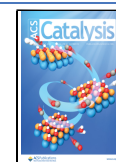
In previous studies, several solid catalysts with Lewis acid and base sites, such as hydroxyapatite (HAP),^{10,11} metal oxides,^{12–14} and $Pd@UiO-66$,¹⁵ have been found to be effective for ethanol upgrading. Among the active catalysts, HAP was particularly noteworthy because of its unusual selectivity for higher alcohols, particularly *n*-butanol (>70% selectivity).^{16–18} Ethanol coupling was proposed to proceed by two different reaction pathways: “Guerbet”^{19,20} and “direct-coupling”.^{21,22} Several studies have addressed that *n*-butanol is formed through a Guerbet reaction including the following four processes: ethanol dehydrogenation, aldolization reaction of acetaldehydes followed by dehydration into 2-butenal, and

subsequent hydrogenation of 2-butenal to *n*-butanol. Others suggested that the *n*-butanol formation process involves a direct surface coupling reaction in which dehydration derives from the OH of one alcohol and the H attached to the α -C of the second.²¹ However, the weak dehydrogenation/hydrogenation over Lewis acid/base sites restrains the ethanol valorization.¹⁹ Considering both perspectives, tailoring the acid–base catalysts by introducing transition metal species (such as Co,²³ Ni,¹³ Cu,^{24,25} Zn,²⁶ and Pd¹⁵) as dehydrogenation sites might improve the performance toward higher alcohols. Cu has been frequently chosen as a dehydrogenation catalyst for ethanol conversion.^{25,27} Recently, Ramasamy and co-workers reported that Cu/Mg_xAl_yO achieved 75% 1-butanol and higher alcohol selectivity at 59% ethanol conversion.²⁵ They found that at a low Cu loading, the Cu^{1+} species is the active site for ethanol dehydrogenation. However, selectively coupling ethanol to C_{6-12} aliphatic alcohols under mild conditions remains an unsolved challenge.

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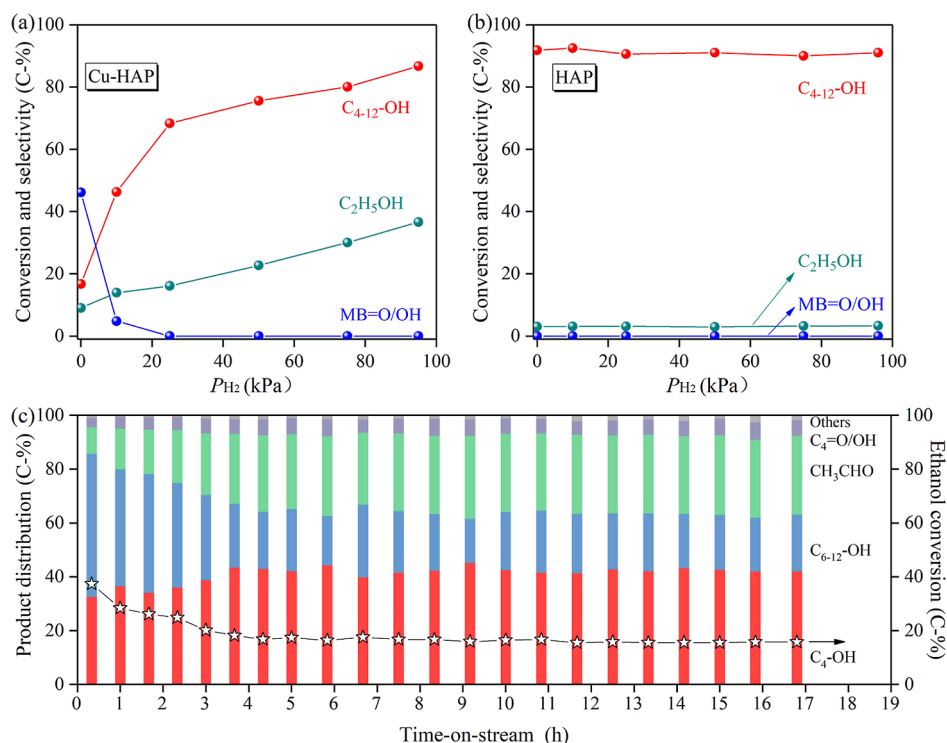


Figure 1. Catalytic performance. Dependence of the product distribution on H_2 partial pressure over 0.5Cu-HAP (a) and HAP (b) catalysts. Reaction conditions: catalyst mass = 100 mg, $P_{C_2H_5OH}$ = 5.7 kPa, balanced with H_2 or N_2 , and the gas flow was 30 mL min⁻¹, ethanol conversion and product distributions observed for the 0.5Cu-HAP and HAP catalysts at 250 °C recorded after 0.5 h on stream. (c) Time-on-stream performance over 0.5Cu-HAP measured at 250 °C. Reaction conditions: catalyst mass = 100 mg, $P_{C_2H_5OH}$ = 5.7 kPa, P_{H_2} = 94.3 kPa, total gas flow rate = 30 mL min⁻¹.

Clearly, unraveling the mechanism of C–C bond coupling during ethanol upgrading would assist us in finding out the key factors that control the product distributions. Recently, Ho et al.¹¹ verified that the Ca–O site catalyzes ethanol dehydrogenation, and Ca–O–P catalyzes condensation of acetaldehyde to form unsaturated 2-butenal in the chain-growth process. This intermediate undergoes cyclization to generate methylbenzaldehydes and methylbenzyl alcohols (MB=O/–OH) on the HAP surface.^{28,29} Besides, 2-butenal can also form *n*-butanol via a Meerwein–Ponndorf–Verley (MPV) reaction using ethanol as the hydrogen donor or undergo a sequence of coupling and MPV steps to form broad distributions of *n*-/iso-alcohols (C_6 – C_{12}).¹⁶ That is, the cyclization of 2-butenal competes with its hydrogenation and coupling reactions.³⁰ As for this consecutive condensation reaction network, controlling the activation of intermediates is a vital factor in enhancing the selectivity of higher alcohols. Inspired by this mechanistic insight, it is possible to tune the ratio of higher alcohols and aromatic alcohols by regulating the hydrogenation of enal intermediates. Modifying the acid–base catalysts with metal nanoparticles might improve the dehydrogenation/hydrogenation capability but unfortunately facilitates unexpected products.³¹ We therefore propose that the cofeeding of H_2 with ethanol might be a potential alternative method to modulate product distribution and increase the selectivity of higher alcohols.

In this study, we report that the Cu-HAP catalyst efficiently converts ethanol to C_4 – C_{12} higher alcohols with a total selectivity of 86.7%, particularly as high as 49.5% selectivity toward C_6 – C_{12} higher alcohols, with 36.6% conversion at 250 °C under a H_2 atmosphere. The HAP support effectively

disperses and stabilizes the small-sized Cu nanoparticles. Because these distinctive metal centers quicken the dehydrogenation rate of ethanol, yielding enough acetaldehyde, the total reaction proceeds to condensation reactions, forming high concentrations of 2-butenal. Subsequently, 2-butenal undergoes a C–C coupling reaction with other aldehydes or direct hydrogenation to produce C_4 – C_{12} higher alcohols. Therefore, it is a feasible strategy to enhance the higher alcohol selectivity via accelerating the ethanol dehydrogenation and intermediate hydrogenation rates over Cu-HAP bifunctional catalysts.

2. EXPERIMENTAL SECTION

2.1. Synthesis. HAP was prepared via coprecipitation, which is based on adding calcium nitrate to diammonium hydrogen phosphate solution and ammonium hydroxide for the adjustment of pH.¹⁷ Cu-HAP was synthesized using the incipient wetness impregnation method. Theoretical amounts of aqueous copper nitrate were added. After impregnation, the samples were kept at room temperature for 2 h and then dried at 50 °C in flowing air. Finally, the samples were calcined at 350 °C in air for 2 h and then reduced at 400 °C for 2 h in 8 vol % H_2/N_2 . The loading amount of the Cu metal detected by inductively coupled plasma optical emission spectrometry (ICP-OES) was about 0.7 wt % (Table S1).

2.2. Characterization. X-ray diffraction (XRD) was performed on a Panalytical X'pert Pro Super X-ray diffractometer with Cu K α radiation (1.5418 Å). High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) was performed using a JEOL JEM-F200. Ex situ X-ray photoelectron spectroscopy (XPS) with Auger electron spectroscopy (AES) was performed using a Thermo

Scientific K-Alpha⁺ spectrometer system equipped with an Al K α X-ray source. Hydrogen temperature-programmed reduction (H₂-TPR) was conducted using a Micromeritics AutoChem II 2920 automated system. In situ ethanol infrared (IR) spectra were obtained using a Bruker 70V spectrometer. In situ diffuse reflectance spectroscopy (DRS)–ultraviolet–visible (UV–vis) spectra were recorded against BaSO₄ on an Agilent Cary 5000 UV–vis–NIR spectrophotometer.

2.3. Catalytic Activity Measurements. Measurements of the reaction were carried out using a fixed-bed quartz-tube reactor with a diameter of 8 mm under atmospheric pressure. Ethanol (5.7 vol %) diluted with H₂/N₂ was passed through the catalyst bed, and the total flow rate was 30 mL min^{−1}. Products were analyzed using an online gas chromatograph with a flame ionization detector (FID). Before tests, methane, as an inert internal standard, was fed into the reactor to bridge the FID and thermal conductivity detector (TCD) for carbon balance calculation. Unless specifically stated, ethanol conversion and product distributions observed for the Cu-HAP catalyst at 250 °C were recorded after 0.5 h on stream. The carbon balance was higher than 98% in each test. The details of the activity tests and calculation methods are given in the Supporting Information.

3. RESULTS AND DISCUSSION

3.1. Effect of H₂ Partial Pressure on Product Distributions in Ethanol Upgrading. Reaction conditions such as atmosphere (H₂ partial pressure) and temperature allow modulation of the product distribution. First, the dependence of the ethanol conversion and product distribution on H₂ partial pressure over Cu-HAP (0.5 wt %) and HAP (Figure 1a,b) was investigated. Figure 1a shows that the ethanol conversion increases significantly when the H₂ partial pressure increases, and an obvious change in product distribution is observed. While the H₂ partial pressure is close to 0, predominant formation of aromatic oxygenates is observed. The selectivity for aromatic oxygenates is as high as 40% over the Cu-HAP catalyst under a N₂ environment, which is in good accordance with our previous research.^{30,32} Meanwhile, additional condensation products are 2-butenal, 2-butenol, and other unsaturated intermediates, resulting from acetaldehyde condensation (Table S2).²⁸ However, a further increase in hydrogen pressure would apparently shift the product selectivity from aromatic oxygenates to higher aliphatic alcohols, and barely any 2-MB-OH is detected when the H₂ partial pressure is above 20 kPa. Notably, Cu-HAP produces nearly 87% higher aliphatic alcohols (37.2% selectivity to *n*-butanol and 49.5% selectivity to C₆–C₁₂ aliphatic alcohols, at 250 °C, 96 kPa H₂, as shown in Table S3), which greatly exceeds the selectivity of the reported ethanol conversion catalysts such as metal oxides and pure HAP.^{10,13,24} Figures 1b and S1 reveal that the ethanol reaction rate is invariant with H₂ partial pressure over the pure HAP catalyst when the H₂ partial pressure is increased from 0 to 95 kPa in the ethanol–H₂ reaction system ($r \sim [\text{H}_2]^0$). Moreover, the selectivities of *n*-butanol, C₆–C₁₂-OH, acetaldehyde, and 2-butenol remain almost unchanged (Figure S2). Our experimental data are in accordance with prior studies reported by Flaherty¹⁶ and Davis.³³ Moreover, there was no detectable deuterated *n*-butanol formed on HAP catalysts when operating in gaseous D₂, which further reveals that hydrogen species originated from ethanol rather than from gaseous H₂.¹⁹ Over HAP, the addition of H₂ has a negligible effect on the ethanol

upgrading process, which is in line with previous research.¹⁶ The ethanol reaction rate follows the zero-order dependence on H₂ partial pressure on HAP, while Cu-HAP has a positive reaction order (Figure S1).³⁴ Although the final products are aliphatic alcohols, these results clearly illustrate that the ethanol reaction pathway is different over Cu-HAP and HAP, and addition of hydrogen does promote the performance of Cu-HAP for ethanol conversion to higher alcohols. The reaction atmosphere has an obvious effect on ethanol condensation, which will be explored and discussed in detail in the following parts.

The temperature dependence of ethanol activity and product distributions on Cu-HAP was explored under different atmospheres. As shown in Figure S3a, under a H₂ atmosphere, higher aliphatic alcohols are the major products below 300 °C. However, the product distributions differ dramatically in a N₂ environment (Figure S3b), and substantial amounts of complex aromatic oxygenates are derived. As such, the formation of aromatic oxygenates is dramatically suppressed below 300 °C when cofeeding H₂. Thus, the addition of sufficient H₂ partial pressure shifts the product distribution from aromatic oxygenates to C₄–C₁₂ higher alcohols via regulating the activation of intermediates over the bifunctional Cu-HAP catalyst.

As shown in Figure 1c, ethanol conversion and product distribution as a function of reaction time on Cu-HAP were further investigated. An initial decrease in conversion from 36.6 to 18% is observed, but insignificant declines in conversion and selectivity are observed for time-on-stream >4 h. The surface restructuring may cause the initial changes in activity for the Cu-HAP catalyst.³⁵ The deposition of large unsaturated compounds could also cover the active center and result in fast deactivation, as has been reported in previous studies.³³ According to thermogravimetric (TG)–mass spectrometry (MS) results, the weight loss starts at 250 °C, and it is negligible during the temperature range of 400–700 °C, indicating soft coke species fouling or covering the active sites (Figure S4).³⁶ After being subjected to calcination, ethanol conversion activity and product selectivity over spent Cu-HAP were nearly restored (Figure S5). In addition, the H₂O molecules produced from the condensation process possibly strongly coordinated to the surface coupling sites, thus leading to the local reconstruction that was likely irreversible under reaction conditions.^{27,37,38} Therefore, we cofed H₂O and ethanol onto pure HAP to gain further insights into the impact of H₂O partial pressure on the reaction performance. The ethanol conversion was reduced as much as possible to reduce the inhibition of water during the dehydration of the intermediates to form *n*-butanol. Obviously, the ethanol conversion rate at 250 °C follows negative order dependence on water partial pressure (~ -0.8) (Figure S6). These results indicate the existence of multideactivation behaviors. Considering our and literature results,²⁰ one can envisage that both coke and in situ produced water can deactivate the active sites.

3.2. Identification of Cu Species. Various characterizations were performed to investigate the physical and chemical features of Cu species and surface chemical properties on these reduced catalysts. In situ XRD and H₂-TPR measurements reflect the structural evolution of copper species under the reduction conditions (Figure S7–9). For the 0.5Cu-HAP catalyst, the measurement was conducted in the temperature range of 50–600 °C under reduction conditions (5% H₂/Ar). No diffraction peaks assigned to copper were

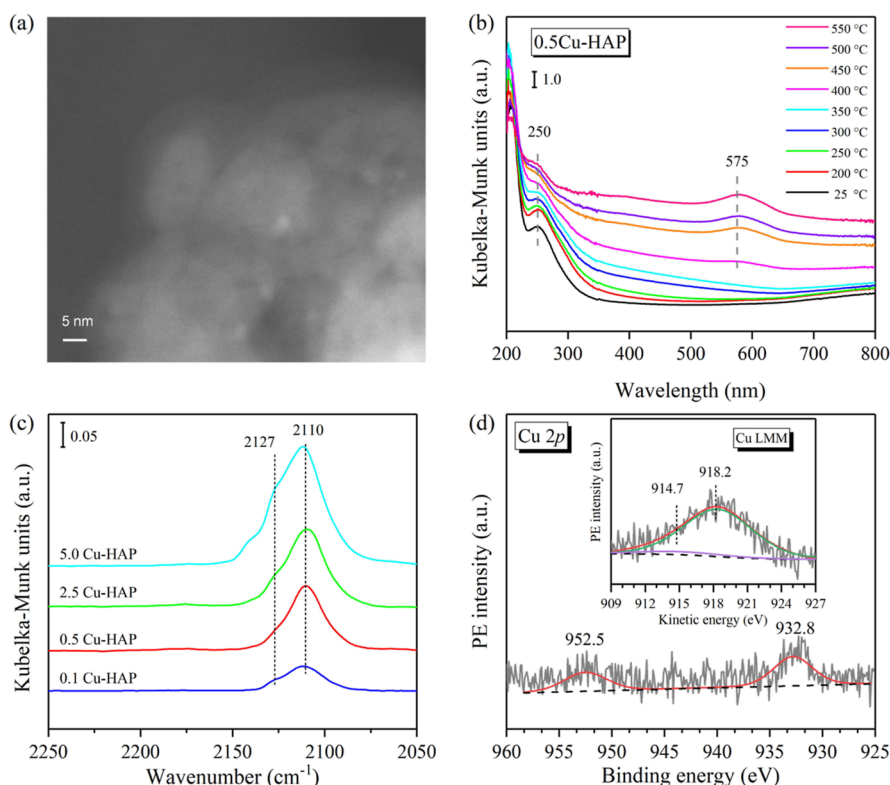


Figure 2. Characterization of Cu-HAP: (a) representative STEM image of reduced 0.5Cu-HAP catalysts; (b) in situ UV-vis DRS over 0.5Cu-HAP catalysts recorded under a 5% H₂/Ar flow; and (c) IR spectra of xCu-HAP, recorded after CO adsorption at 25 °C. (d) XPS and Cu LMM Auger spectra of 0.5Cu-HAP catalysts.

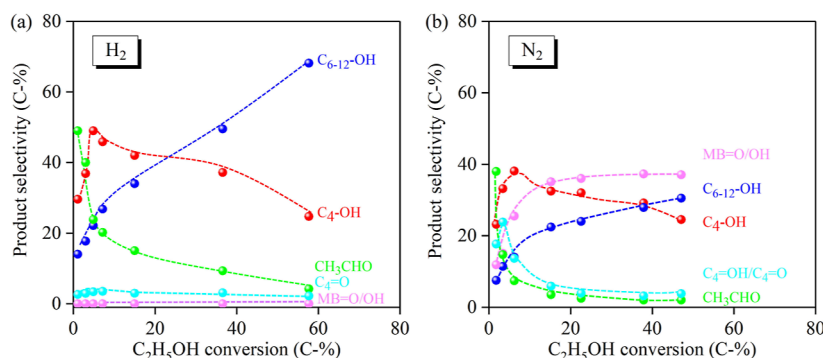


Figure 3. Dependence of the product distributions as a function of ethanol conversion over 0.5Cu-HAP under a (a) H₂ atmosphere and (b) N₂ atmosphere. The dashed lines serve as a guide. Reaction conditions: $T = 250\text{ }^{\circ}\text{C}$, $P_{\text{C}_2\text{H}_5\text{OH}} = 5.7\text{ kPa}$, balanced with H₂ or N₂, the gas flow was 30 mL min^{-1} , and $\text{WHSV} = 1.0\text{--}40\text{ g}_{\text{C}_2\text{H}_5\text{OH}}\text{ g}_{\text{cat}}^{-1}\text{ h}^{-1}$, ethanol conversion and product distributions observed for the 0.5Cu-HAP catalyst at 250 °C recorded after 0.5 h on stream.

detected even at 600 °C. This result implies that either amorphous or finely dispersed Cu species were formed within Cu-HAP with low Cu contents. Further HAADF-STEM images show that small metallic copper particles of 2–3 nm were observed on the reduced 0.5Cu-HAP sample (Figures 2a, S10, and S11), in which the white dots are definitely assigned to Cu particles due to the Z-contrast distinction.³⁹ For catalysts with higher Cu loading ($\geq 2.5\%$), the STEM images show an average Cu particle diameter of 4.9 nm for 2.5Cu-HAP and 13.8 nm for 5.0Cu-HAP (Figures S12 and S13 and Table S1). UV-vis spectra obtained in the diffuse reflectance mode of all prepared xCu-HAP samples exhibit a strong peak at 250 nm, and the peak was attributed to the O²⁻ $\rightarrow$ Cu²⁺ charge transfer transition (Figure S14).⁴⁰ Moreover, in situ

UV-vis DRS was conducted to probe the electronic structure evolution process (Figure 2b). Upon the reduction of 0.5Cu-HAP at 400 °C under a 5% H₂/Ar flow, the Cu²⁺ ions were reduced (intensity of the peak at 250 nm), and a band at 575 nm assigned to metallic copper gradually emerged.^{41–43} The corresponding peak intensity increased clearly upon increasing treatment temperature higher than 400 °C.

Furthermore, the electronic states of Cu species were investigated using CO-diffuse reflectance IR Fourier transform (DRIFT) spectroscopy. The spectra were collected and are shown in Figures 2c and S15. Before the measurement, the corresponding catalyst was reduced in 8% H₂/N₂ mixture gas at 400 °C for 1 h. The reduced 0.5Cu-HAP sample shows two main adsorption bands centered at ca. 2110 and 2127 cm⁻¹,

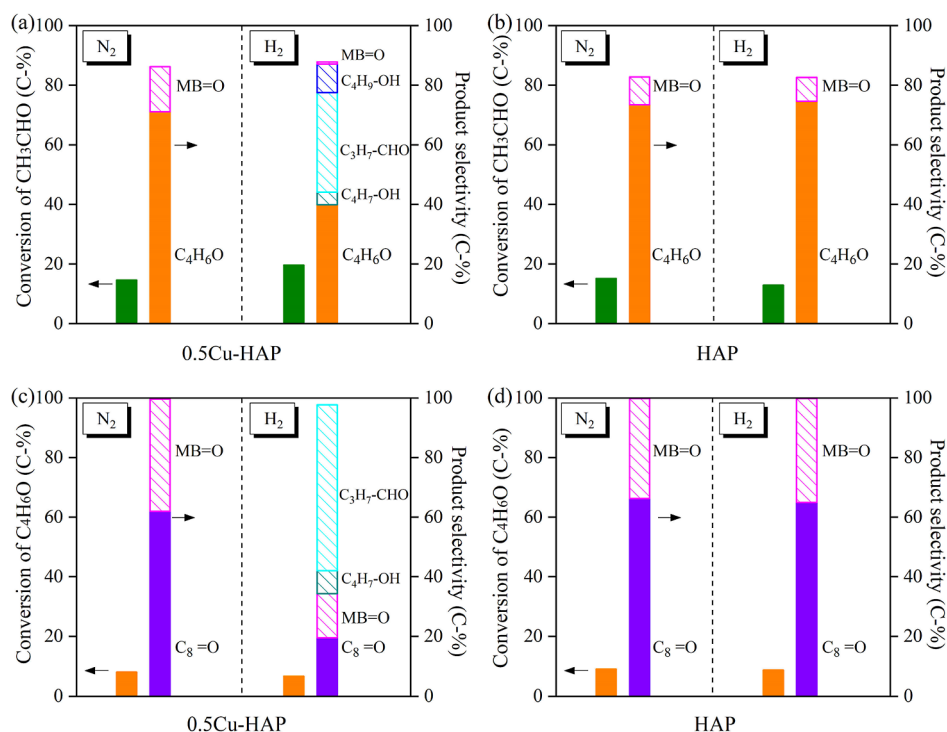


Figure 4. Comparison of intermediate aldehyde reaction conversion and selectivities over 0.5Cu-HAP and HAP at 250 °C upon feeding (a,b) 1 vol % CH₃CHO balanced with N₂ or H₂ and (c,d) 1 vol % C₄H₆O balanced with N₂ or H₂.

respectively. The former is assigned to CO adsorbed on the interfacial Cu nanoparticles with small sizes,^{27,39,44} and the latter probably corresponds to CO linear adsorption on Cu⁺ species.^{45,46} Note that only a strong chemical interaction between the metal and support could result in these distinctive metal structures. Thus, these highly dispersed Cu nanoparticles presented on the reduced 0.5Cu-HAP catalyst surface allow it to catalyze the ethanol dehydrogenation and aldehyde hydrogenation reactions. XPS analysis was performed to further disclose the chemical valence of Cu species. The Cu 2p spectrum shows a peak at ca. 932.8 eV, which corresponds to Cu⁰ and/or Cu⁺ species on HAP^{47,48} (Figure 2d). Two overlapping peaks in the Cu LMM spectrum with a maximum at 918.2 and 914.7 eV further indicate the existence of Cu⁰ and Cu¹⁺ species,⁴⁹ respectively (Figure 2d), which is consistent with the result of CO-absorption Fourier transform IR (FTIR) spectra and in situ UV–vis DRS.

3.3. Insights into the Reaction Pathways. The changes in product selectivity with increasing ethanol conversion were investigated under different atmospheres in order to elucidate the possible ethanol conversion network on Cu-HAP. As shown in Figure 3a, selectivity extrapolated to near-zero conversion shows that acetaldehyde (C₂H₄O) is the sole primary product on Cu-HAP, obtained directly from ethanol dehydrogenation. With an increase in ethanol conversion, the selectivity of acetaldehyde gradually decreases. Subsequently, 2-butenal (C₄H₆O) forms through the self-coupling of acetaldehyde,^{9,28,29,50} however, the gas concentration of 2-butenal is much lower than that of butyraldehyde and *n*-butanol, suggesting that the hydrogenation reaction of 2-butenal takes place more easily. The butyraldehyde selectivity quickly reaches a plateau when the formed butyraldehyde is immediately hydrogenated to *n*-butanol, and the needed hydrogen species are derived from either ethanol or gaseous hydrogen.¹³ The selectivity of *n*-butanol increases to 45% at

10% conversion and then decreases at a higher conversion. Meanwhile, the selectivity toward C_{6–12} alcohols and the average carbon number of condensation products increase with the increasing conversion of ethanol. The product distribution shows that ethanol condensation follows the step-growth kinetic model.¹⁶ Also, nearly no aromatic aldehydes/alcohols are detected in the reaction. These findings illustrate that the unsaturated intermediate aldehydes are hydrogenated to saturated aliphatic alcohols under a H₂ atmosphere instead of cyclization to aromatic oxygenates.

To shed light on the reaction pathway, changes in product selectivity with increasing ethanol conversion under a N₂ atmosphere are shown in Figure 3b. Under a N₂ atmosphere, acetaldehyde is also the primary product, but both aromatic oxygenates and aliphatic alcohols are the final outcomes.³⁰ Thus, the reaction of 2-butenal hydrogenation and condensation to C₄–C₁₂ aliphatic alcohols competes with its aromatization reactions to MB=O/-OH (producing C₈ aromatic oxygenates).

Furthermore, Figure S16 shows the MB=O(OH)/C₄–C₁₂OH selectivity ratio for a range of ethanol conversions (1–60%). It is observed that under a H₂ atmosphere, the ratio is about 0, with ethanol conversion ranging from 1 to 60%. As for measurement investigated under a N₂ atmosphere, the MB=O(OH)/C₄–C₁₂OH ratio increases from 0.38 at low ethanol conversion to 0.65 at about 40% ethanol conversion. The experimental data show that hydrogenating intermediate aldehydes to saturated products is not significant under an inert atmosphere due to the low concentration of hydrogen species. Both the reaction atmosphere and ethanol conversion have impact on the product distribution, and the atmosphere has a critical impact on controlling the reaction route.

To disclose the reaction routes and verify the vital impact of the H₂ atmosphere on controlling the product distribution, kinetic measurements of intermediates were investigated via

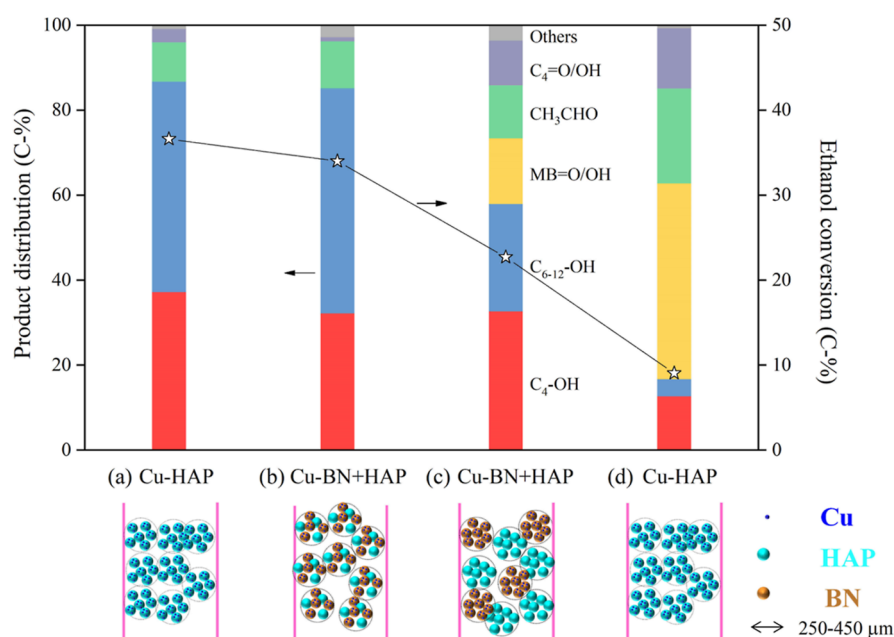


Figure 5. Influence of the proximity of Cu species and HAP on catalytic behaviors during ethanol conversion. (a) 0.5Cu-HAP prepared by the incipient wetness impregnation method, sieved to 40–60 mesh. (b) Cu/BN and HAP powders were finely ground together and sieved to 40–60 mesh. (c) Stacking of 40–60 mesh Cu/BN and 40–60 mesh HAP. (d) 0.5Cu-HAP prepared by the incipient wetness impregnation method, sieved to 40–60 mesh. All catalysts (Cu and HAP species mass ratio = 1:200) were evaluated at 250 °C, and the gas flow was 30 mL min^{−1}. Ethanol conversion and product distributions observed for Cu-HAP composite catalysts at 250 °C recorded at the start of reaction (0.5 h).

feeding acetaldehyde and 2-butenal onto Cu-HAP or HAP catalysts under a N₂ or H₂ atmosphere. As seen in Figure 4a, 2-butenal and methylbenzaldehydes are evidently produced when acetaldehyde is the sole feed under a N₂ atmosphere, with barely any hydrogenation products observed. Notably, the selectivity to hydrogenated products was >45% (~35% as butyraldehyde) in the direct reactions of acetaldehyde under a H₂ atmosphere, with only traces of aromatic oxygenates (less than 1%). According to previous reports, retaining the C=C bond is of great importance for ethanol condensation to aromatic oxygenates.³⁰ The result demonstrates that Cu-HAP does catalyze the C=C bond hydrogenation and favors the production of butyraldehyde. Correspondingly, under higher hydrogen partial pressures, C=C bonds generated by the aldol condensation reaction could not be easily retained over the Cu-HAP catalyst. As shown in Figure 4b, the reaction atmosphere had no observable effect on the main product distribution, demonstrating that pure HAP cannot catalyze intermediate hydrogenation under a H₂ atmosphere. These observations are consistent with previous measurement results, suggesting that H₂ has neither a kinetic nor thermodynamic effect during the ethanol condensation process.

Subsequently, 2-butenal was employed as a reactant to elucidate the formation pathways of aliphatic alcohols and aromatic oxygenates on Cu-HAP under different atmospheres (Figure 4c). Catalytic results show that under a N₂ atmosphere, linear 2,4,6-octatrienal (C₈ trienal), formed by self-coupling of 2-butenal, is the primary product, which sequentially produces ortho-aromatic oxygenates as secondary products. Under a H₂ atmosphere (Figure 4c), the main product is butyraldehyde (>50%), while the total selectivity of 2,4,6-octatrienal and ortho-aromatic oxygenates is <35%, which indicates the preferential C=C bond hydrogenation of 2-butenal over the Cu-HAP catalyst.^{24,51,52} This result indicates that in a H₂ atmosphere, a hydrogen-rich surface is

formed on Cu-HAP, which enhances the formation of higher alcohols, restraining the production of unwanted cyclization products. On the contrary, no hydrogenation products are observed when 2-butenal is cofed with H₂ over the HAP catalyst (Figure 4d).^{11,18} Comparing the acetaldehyde/2-butenal conversion over Cu-HAP and HAP catalysts under a H₂ atmosphere indicates that copper species contribute to H₂ activation to form a hydrogen-rich surface and the H-transfer ability is enhanced. In detail, it can be seen that the copper species has a key impact on facilitating C=C bond hydrogenation of 2-butenal to produce butyraldehyde, making this a distinctive reaction route from ethanol to aromatic oxygenates.⁵²

3.4. Revealing the Function of H₂. Compared with an inert atmosphere, H₂ not only increased the conversion rates of ethanol over Cu-HAP but also considerably promoted the production of C₆–C₁₂ higher alcohols, indicating that the synergistic effect between hydrogen and Cu-HAP plays a vital role in modulating the reaction pathway. To further explain the promoting role of H₂ and Cu species, as well as correlate the catalyst structure and activity, extensive kinetic measurements were carried out. Furthermore, the ethanol conversion shows a volcano-type catalytic behavior with the Cu content on HAP increasing from 0.1 to 5.0 wt %. For the blank HAP catalyst, the ethanol conversion is below 3% at 250 °C, and *n*-butanol is the main product. The selectivity toward *n*-butanol and C₆–C₁₂ alcohols is 65.6 and 26.1%, respectively. A relatively lower copper content on HAP is more favorable for the production of higher alcohols, which is in agreement with previous studies done by Huber and co-workers²⁷ and others.²⁵ With the increment of metal content, the conversion of ethanol gradually increases to a plateau ca. 40% (Figure S17). The selectivity of C₆–C₁₂ higher alcohols also obviously increased (~50%). However, a further increase in the copper content led to a lower conversion of ethanol (19% for 5Cu-HAP). Notably,

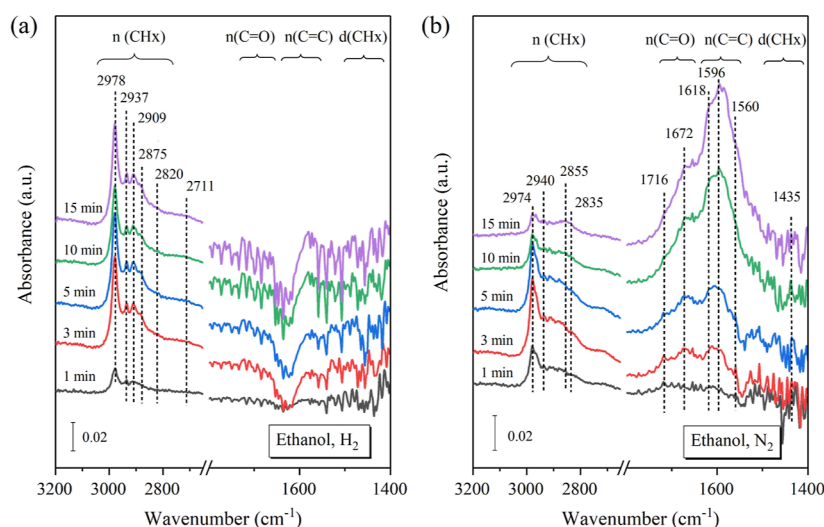


Figure 6. In situ DRIFT spectra recorded over 0.5Cu-HAP under a (a) H_2 atmosphere at 200 °C and (b) N_2 atmosphere at 250 °C. Ethanol (3×10^{-7} mol) was dosed onto the cell, and then, the spectra were recorded.

a decreased selectivity toward *n*-butanol and C_6 – C_{12} alcohols was observed because of the remarkable increase in the selectivity to acetaldehyde (>50%), indicating that the HAP catalyst with a higher copper content retarded the C–C coupling process.^{53,54} Considering our previous results and the knowledge from literature, the 5Cu-HAP catalyst with the largest copper nanoparticles was inferior to the catalysts with lower Cu contents, revealing that enal intermediate hydrogenation was sensitive to the Cu particle size.^{55,56} The activation of hydrogen also has a great impact on determining the product distribution. For example, the 0.5Cu-HAP sample results in the darkest color of tungsten species during the H_2 reduction process (Figure S18), while the color was unchanged over WO_3 alone. The highly dispersed copper nanoparticles and migration of hydrogen on the HAP surface may contribute to the development of such a hydrogen-rich surface. The increase in surface hydrogen concentration promotes the hydrogenation process, thus enhancing the higher alcohol selectivity.^{27,54} These results indicate that the “synergistic effect” occurred between copper and HAP and the addition of hydrogen changes the reaction route of ethanol.

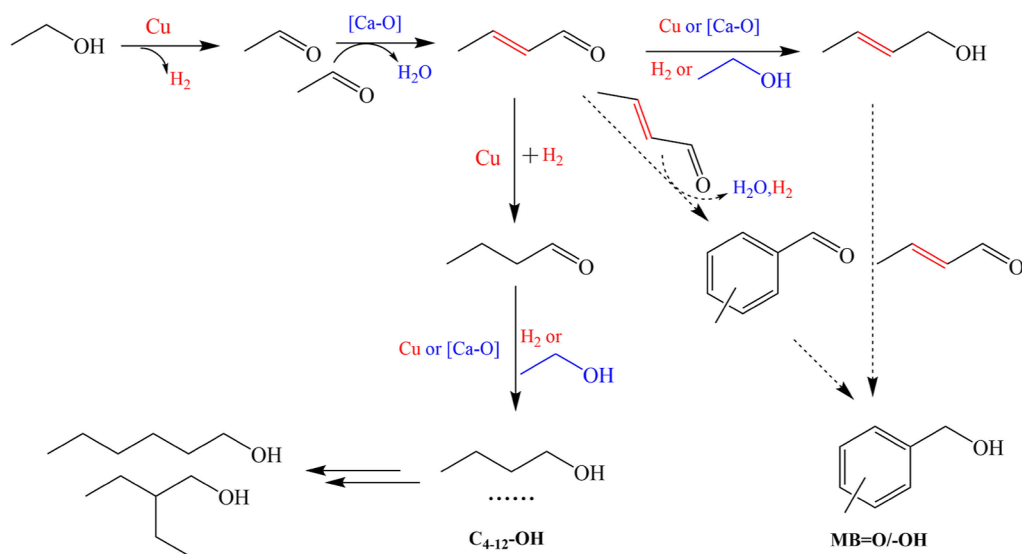
By decoupling the ethanol reaction network, the proximity effect of Cu species and HAP on the activity and selectivity was further explored. Three kinds of Cu-based and HAP composite catalysts with different distances between Cu species and HAP were prepared by different preparation methods. Catalytic results of these catalysts at 250 °C are shown in Figure 5. For the first two integration methods (Figure 5a,b), the C_4 – C_{12} alcohol selectivities were above 85%, with only traces of aromatic oxygenates found under a H_2 atmosphere (less than 0.5%). Our recent work showed that Cu/BN could effectively catalyze ethanol dehydrogenation to acetaldehyde.⁵⁷ The Cu/BN and pure HAP powders that were directly mixed showed similar activity and selectivity to the Cu-HAP (0.5 wt %) catalyst, demonstrating that the atomic distance between copper species and HAP is not considered critical.²⁷ Then, we mixed the granules of Cu/BN and HAP together in order to further increase the distance of two components (Figure 5c). Even under a H_2 atmosphere, small amounts of aromatic oxygenates with a selectivity of 15.5% were formed due to the quick cyclization of 2-butenal, and the C_4 – C_{12} alcohol selectivity decreased accordingly (<60%). As the distance of

Cu species and HAP decreases, the yield of C_4 – C_{12} alcohols increases substantially. Preliminary results suggest that copper species selectively catalyze alcohol dehydrogenation reactions and promote the 2-butenal intermediate hydrogenation to butyraldehyde, while calcium oxide/phosphate sites are active centers for the C–C bond formation. As shown in Figure 5, the proximity between Cu and HAP plays a key role in producing higher alcohols and restraining the unwanted cyclization reactions. A smaller distance between Cu species and HAP will promote the rapid transfer of 2-butenal molecules from HAP to copper species, thus enhancing the selective hydrogenation of 2-butenal to butyraldehyde on the surface of copper nanoparticles under a H_2 atmosphere. Though bifunctional active sites contacted sufficiently, the 0.5Cu-HAP catalyst exhibited much higher C_8 aromatic oxygenate selectivity (46.1%) and lower ethanol conversion (22.3%) due to the insufficient hydrogen partial pressure provided by the ethanol dehydrogenation process (Figure 5d). In order to further clarify the ethanol condensation process under different atmospheres, in situ UV–vis analysis was conducted (Figure S19). Under a H_2 atmosphere, the band at 575 nm was clearly observed, indicating that a reduction atmosphere allows Cu nanoparticles to maintain the metallic state.

To gain a deeper insight into the higher alcohol and aromatic oxygenate formation mechanism on Cu-HAP under H_2 and N_2 atmospheres, in situ DRIFT experiments were conducted (Figure 6a,b). Under a H_2 atmosphere, a series of typical IR bands at 2978, 2937, 2909, and 2875 cm^{-1} gradually increased, demonstrating the fast coupling of ethanol to higher alcohols.^{22,58} Under a N_2 atmosphere, two weak bands at 1716 and 1672 cm^{-1} , which were attributed to the $\nu(\text{C}=\text{O})$ modes of acetaldehyde and 2-butenal, respectively, were observed in the first 3 min.⁵⁹ Especially, the main band at 1596 cm^{-1} gradually emerged, which was attributed to characteristic features of C_8 aromatic oxygenates.³⁰ The IR results are consistent with the above-mentioned kinetic analyses.

Based on our spectroscopic investigations and kinetic measurement of feeding ethanol or intermediates (acetaldehyde, 2-butenal) under a H_2 or N_2 atmosphere, we summarized the reaction pathway of ethanol conversion to C_4 – C_{12} alcohols and aromatic oxygenates over Cu-HAP

Scheme 1. Reaction Network of Ethanol Upgrading to Higher Alcohols and Aromatic Oxygenates on Cu-HAP under a H₂ Atmosphere (Solid Arrows) and N₂ Atmosphere (Dashed Arrows)



(Scheme 1). Ethanol is first dehydrogenated to acetaldehyde over Cu sites.⁶⁰ Then, the self-condensation and dehydration reaction of acetaldehyde produces 2-butenal over an acid–base active center. Under a H₂ atmosphere, the Cu site primarily leads to the hydrogenation of the C=C bond of 2-butenal to produce butyraldehyde, which then undergoes hydrogenation to *n*-butanol.¹⁶ Other intermediate aldehydes undergo self- or cross-condensation and subsequent hydrogenation to produce C₆–C₁₂ higher alcohols. No trace of aromatic oxygenates were detected under a H₂ atmosphere even at high ethanol conversion levels. In a N₂ atmosphere, aromatic oxygenates are formed through the self-coupling of 2-butenal or cross-condensation of 2-butenal with 2-butenol.^{29,30} Detailed investigations revealed that the addition of hydrogen was the critical parameter to promote the shift of products from aromatic oxygenates to aliphatic alcohols.

4. CONCLUSIONS

We report a new catalytic system whereby bifunctional Cu-HAP can facilitate valorization of ethanol to higher alcohols under a H₂ atmosphere. At 250 °C, the total C₄–C₁₂ alcohol selectivity over Cu-HAP reached 86.7%, and ethanol conversion was above 36.6%. According to kinetic analysis, a higher ethanol conversion is achieved with hydrogen, showing a positive order dependence on the H₂ partial pressure. The rate and product distribution remained constant on HAP with increasing H₂ partial pressure, which demonstrates that ethanol condensation proceeds according to the “Guerbet” mechanism over Cu-HAP and a “direct-coupling” route over pure HAP, respectively. A combination of kinetic experiments discloses that aromatic oxygenates are produced via ethanol dehydrogenation, aldol condensation, and dehydrocyclization reactions, and during the process, 2-butenal is the crucial intermediate. Addition of hydrogen contributes to the hydrogenation of C=C and C=O bonds of 2-butenal to form saturated *n*-butanol over the Cu-HAP catalyst, further regulating the reaction pathway. The experiments of the catalyst integration methods further unravel that Cu sites favoring the C=C bond hydrogenation are of great importance for shifting the reaction pathway. Overall, these

results provide a fundamental understanding on how the intermediate hydrogenation reaction pathway can be fine-tuned to optimize the product selectivity for biomass-based ethanol valorization.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.2c03327>.

Catalyst preparation; characterization methods; catalyst evaluation; dependence of the ethanol reaction rate on H₂ pressure over HAP and 0.5Cu-HAP; dependence of the product distribution on H₂ partial pressure over HAP; dependence of the product distribution on the reaction temperature over 0.5Cu-HAP; TG–MS profiles of spent 0.5Cu-HAP; stability and regeneration tests of Cu-HAP; effect of H₂O addition in the feed on the reaction rate of ethanol; XRD patterns of xCu-HAP; in situ XRD patterns of 0.5Cu-HAP; H₂-TPR profiles of xCu-HAP; TEM images of 0.5Cu-HAP; STEM images of reduced 0.5Cu-HAP, 2.5Cu-HAP, and 5.0Cu-HAP; UV–vis spectra of calcined xCu-HAP; in situ CO adsorption and desorption DRIFT spectra over reduced xCu-HAP; effect of reaction atmosphere on the condensation compound selectivity ratio; performance of ethanol on xCu-HAP; photographs of WO₃ mixed with Cu-based catalysts; in situ UV–vis DRS spectra during the ethanol reaction over 0.5Cu-HAP; structure parameters of xCu-HAP catalysts; distribution of major products from ethanol over the 0.5Cu-HAP catalyst; and summary of the catalytic data of some representative catalysts used in ethanol coupling reactions (PDF)

■ AUTHOR INFORMATION

Corresponding Author

An-Hui Lu – State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, China; orcid.org/0000-0003-1294-5928; Email: anhuilu@dlut.edu.cn

Authors

Bai-Chuan Zhou — State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, China

Wen-Cui Li — State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, China; orcid.org/0000-0001-8066-7144

Wen-Lu Lv — State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, China

Shi-Yu Xiang — State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, China

Xin-Qian Gao — State Key Laboratory of Fine Chemicals, Liaoning Key Laboratory for Catalytic Conversion of Carbon Resources, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, China

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acscatal.2c03327>

Author Contributions

B.-C.Z., W.-C.L., and A.-H.L. conceived and designed the project. B.-C.Z. performed the catalytic experiments, characterization, and spectroscopic investigation; W.-L.L. and S.-Y.X. helped conduct kinetic and structure characterizations. X.-Q.G. helped perform the STEM measurements. A.-H.L. revised the manuscript. All authors have reviewed and contributed to the published version of the manuscript.

Notes

The authors declare no competing financial interest.

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